Essentialism and the Essence of what it is to be Necessary

Fine (1994) argued against the view that essence should be understood in terms of *de re* modality. He instead suggested understanding metaphysical necessity by the essence of things. The idea to explain, especially to ground all metaphysical necessity in essence was followed by various philosophers albeit with some differences in details. I term such accounts "*essentialism on the source of necessity*", or simply "*essentialism*".

However, if essence is not to be understood in terms of modal notions, one might ask what makes essence so intimately related to necessity. Some philosophers argue that essentialists cannot account for why *what it is to obtain essentially* is such that all facts obtaining essentially are necessary. If essence is partially grounded in necessity, essentialism violates the asymmetry of ground; but if what it is to obtain essentially should not be understood in terms of what it is to be necessary, then there is a conceptual gap between what it is to obtain essentially and what it is to be necessary.

In this paper, I argue that there is no conceptual gap between the non-modal conception of essence and necessity, if one takes metaphysical necessity to be essentially essentialist. After some terminological clarifications in §1, §2 is devoted to elaborating the gap challenge. In §3, I first argue that essentialists must take the extensional relation between essence and necessity—that every fact obtaining essentially is necessary—to obtain essentially of *what it is to be necessary*. Then, I proceed to show that the gap challenge can be answered by appealing to the supposed nature of *what it is to be necessary*. In §4, I argue that my proposal is not viciously circular.

1 Background

1.1 Terminology

By "necessity" I mean *metaphysical* necessity unless otherwise indicated. "$\Box$" will be exclusively used as an operator for metaphysical necessity, corresponding to the natural language predicate "*to be metaphysically necessary*". Similarly, if there are varieties of grounding, by "grounding" I mean *metaphysical* grounding. By "essence", I mean *mediate and consequential* essence, following Fine (1995a,c). Suppose we have a clear idea of a strict and direct essence, which is immediate and constitutive. Then, the mediate essence of F includes both the essence of F itself and the essence of things that are involved in the constitutive essence of F; the consequential essence of F includes what logically follows from the nature of F.

Also crucial is the distinction between objectual essence and generic essence. On the one hand, there is objectual essence, the essence of a particular object; and on the other hand, there is generic essence, the essence of *what it is to be* such-and-such. Fine (2015) argues that the generic essence of F is different from the objectual essence of the concept F. Suppose the objectual essence of the concept *concept* includes the fact that *concept* is a concept, and therefore it is consequentially essential that $\exists x(x \textit{ is a concept})$. However, it seems quite implausible to think that it is by the nature of what it is to be a concept that there is something satisfying it. That is, while $\exists x(x \textit{ is a concept})$ is plausibly essential to the concept *concept*, it would not be essential to *what it is to be a concept*. This, we may say, illustrates the difference between the objectual essence of the concept *concept* and the generic essence of what it is to be a concept.

I do *not* take "to be essential to" and "to obtain essentially of" (equivalent to "to obtain in virtue of the nature of") to be interchangeable, since I operate with the non-factive notion of essence, under which facts essential to something[1] may not obtain. If a fact obtains essentially of something, it is essential to the thing in question; but not vice versa. Following Fine (2015), by $\phi \leftarrow_{x_1,\ldots,x_n} F(x_1,\ldots,x_n)$ and $\psi \leftarrow \chi$, I mean ϕ is generically (mediately and consequentially) essential to F and ψ is generically essential to χ respectively. I take generic essence to

[1]Expressions like "something", "whatever", "entities", etc. should not be taken to include only first-order objects.

be primary, and objectual essence to be a subclass of generic essence. Objectual essence will be notated by using a *rigid predicate* $\lambda x.(x = a \vee x = b \vee \dots)$, where $a, b, \dots$ are names for objects. Thus, to say ϕ is objectually essential to a, I write "$\phi \leftarrow_x x = a$". For the notion of obtaining essentially of something, I borrow from the traditional notation[2] and write "$E_F\phi$" to mean ϕ obtains essentially of F, where "F" is possibly 0-placed. By convention, I sometimes write "$E_a\phi$" as an abbreviation of "$E_{\lambda x.x=a}\phi$". I endorse the notation for generic grounding from Fine (2015), where $F(x_1,\dots,x_n) \rightarrow_{x_1,\dots,x_n} G(x_1,\dots,x_n)$ means F generically grounds G. Here the grounding is also non-factive.

By "an essence fact" I mean a fact of the form $\phi \leftarrow \psi$. Furthermore, I assume that if ϕ is essential to F, this essence fact obtains essentially of F. That is, if $\phi \leftarrow_{x_1,\dots,x_n} F(x_1,\dots,x_n)$, then $E_F(\phi \leftarrow_{x_1,\dots,x_n} F(x_1,\dots,x_n))$. By "a realized essence fact" I mean a fact of the form $E_F\phi$. I also assume that a realized essence fact obtains essentially: $E_F\phi \supset E_F E_F \phi$, where "$\supset$" is used for the material conditional to prevent ambiguity.

I take the primary subjects of being necessary and obtaining essentially to be facts. Nevertheless, it should be straightforward to translate the terminologies used in this paper to make them conform to other favored theories. Sometimes I will use "an essence of F" to denote a fact essential to F, while reserving "nature" strictly for the non-further-entity sense of essence (Lowe, 2008).

Finally, I introduce the predicate "ε", which corresponds to the natural language predicate "to obtain essentially of something". "εp", then, means that p is a fact obtaining essentially.

1.2 Essentialism on the source of necessity

Throughout the discussion, I will focus on a strong form of essentialism on the source of necessity: namely, the thesis that *to be necessary* is grounded in *to obtain essentially* (viz., $\varepsilon p \rightarrow_p \Box p$). Except where it might cause ambiguity, by "*essentialism on the source of necessity*", or simply "*essentialism*, I will denote the strong thesis above. Essentialism, in this sense,

[2]Yet I do not endorse Fine (1995a)'s logic of essence wholeheartedly, since the axiom (II) (v): $F \subseteq G \supset (E_F\phi \supset E_G\phi)$ yields the metaphysically undesirable result that for any two intensionally equivalent predicates F and G $E_F\phi$ iff $E_G\phi$, regardless of their hyperintensional difference.

would yield that for any necessary fact p, $\Box p$ is fully grounded in the fact that εp. This can be regarded as a proposed account of the ultimate metaphysical source of necessity (Hale, 2002). The source of necessity is supposed to provide the metaphysical explanation of why a necessary fact is necessary. It seems unlikely that a source of necessity must always also explain why the fact obtains.[3] Sometimes ϕ may occur in explaining the necessity of ϕ. It is customary to explain the necessity of the identity between Hesperus and Phosphorus by their actual identity and the necessity of identity.[4] In this case, the particular source for the necessity of ϕ would include ϕ, and thereby be inadequate to explain ϕ itself. Yet this cannot be the *ultimate* source for necessity, for in explaining the necessity of the fact that Hesperus is Phosphorus we appealed to the necessity of identity. We might further ask why, then, every identity between an object and itself is necessary; and if it is again explained by appealing to the necessity of another fact ψ, we might ask why ψ is necessary, and so on.

The chain of explanation might continue infinitely, or end at the point where we can give no further explanation for the necessity in question, i.e., where the necessity is primitive. Finally, the chain might end if one could provide an explanation for the necessity in question with a non-modal source. The final non-modal terminus would then be the *ultimate* source of the particular necessity which was initially at stake.

Essentialism, as I understand it, then, implies the following two theses: that the ultimate sources of *all* necessities are some realized essence facts, and that the source of necessity is the ground of the necessity. The first thesis is particularly strong. Some philosophers favor a weaker view: that some necessities have their ultimate sources in essence facts, but some other necessities do not[5]. In this paper, I focus on the more ambitious essentialism which holds that all necessities are grounded in realized essence facts.

Although the grounding thesis is explicitly endorsed by some philosophers (e.g., Lowe, 2012; Hale, 2013; Wallner and Vaidya, 2020), Correia (2024) suggests that the grounding thesis is not forced upon theorists who wish to seek the ultimate source of necessity in essence. He suggests

[3] Bovey (2022) argues that essentialism fails if every source of necessity must also explain why the fact obtains.

[4] Kripke (1980). Although it might be disputable whether this is a *metaphysical* explanation, at least presumably there is no quick argument against it.

[5] For example Hale (2013, Ch.6) holds that the necessity of an objectual essence fact is primitive, and Godman et al. (2020), Correia and Skiles (2021) doubt that the necessity of logical truths has an essentialist source.

that instead of grounding, one might hold that necessity is reduced to, or defined in terms of essence (as in Fine (1994, 1995a)), or that there is a generic identity between essence facts and necessities (as in Correia and Skiles (2021)). I admit that these are real possibilities, and by "essentialism" I *exclude* accounts that do not involve the grounding thesis.[6] Nevertheless, it should be noted that there indeed are cases where the grounding thesis is implicit. For example, Kit Fine, when defining *to be necessary* in terms of *to be essential to all objects* arguably has a real constitutive definition in mind, which gives both grounding and essence claims (Fine, 2015).

Finally, some comments should be made on the occasional restriction to *objectual* essence. This qualification is explicit in some writings, for example Fine (1994, 9) writes that: "[t]he metaphysically necessary truths can then be identified with the propositions which are true in virtue of the nature of all *objects* whatever" (italics added). Similarly, Hale (2013, 150) notes that "[m]etaphysical necessities might plausibly be identified as those which hold in virtue of the nature of *things* in general" (italics added).

Essences are straightforwardly not always necessary in paradigms where the essence of what it is to be something is treated as a constitutive necessary condition (Fine, 2015), or as something generically identical with it (Correia and Skiles, 2017). For example, let "*sg*" stand for the fact that *snow is green*. Then it is extremely plausible that $sg \leftarrow sg$; yet certainly it is not necessary that snow is green. After all, it does not even actually obtain. This might seem to provide a motivation for the restriction to objectual essence, since we clearly cannot license $\Box sg$. However, $sg \leftarrow sg$ simply does not grant εsg in our sense. As I mentioned earlier, the *essence fact* about F — not the essence of F — obtains essentially of F. Thus, $sg \leftarrow sg$ licenses only $E_{sg}(sg \leftarrow sg)$. And since $p \leftarrow q$ yields $q \supset p$ as Fine (2015, 299) suggested, we could consequentially infer $E_{sg}(sg \supset sg)$, and, of course, $\varepsilon(sg \supset sg)$.[7] This is acceptable, since we would have no problem accepting $\Box(sg \supset sg)$: necessarily, snow is green if snow is green. I would like to take one step further and contend that $E_{sg}(sg \leftarrow sg)$ is *grounded* in $sg \leftarrow sg$, yet

[6] For instance the account which holds that facts that are necessary and obtaining essentially are extensionally equivalent, yet necessity is ungrounded (Wilsch, 2017). Correia and Skiles (2021) also hold that *to be necessary* is generically identical with *to be a logical consequence of an essence fact* (or its extensional correlate), which is incompatible with the grounding thesis.

[7] In general, $p \leftarrow_{x_1,\ldots,x_n} F(x_1,\ldots,x_n)$ would yield $\forall x_1 \ldots \forall x_n (F(x_1,\ldots,x_n) \supset p)$.

this is optional and perhaps contentious.

Thus, I see no particular reason for essentialists to think that only objectual realized essence facts can ground necessity. It is even unclear how an essentialist would account for some necessities without appealing to generic essence. For example, consider the necessary fact that *all bachelors are unmarried*. Let "U" be the predicate "*to be unmarried*" and "B" "*to be a bachelor*". With generic essence we can have $Ux \leftarrow_x Bx$, which yields $E_B \forall x(Bx \supset Ux)$. This would ground $\Box \forall x(Bx \supset Ux)$, i.e., it is necessary that all bachelors are unmarried. But it is not clear at all how an essentialist could account for this necessity by appealing only to objectual essence. Certainly, it is not by the nature of all bachelors that they are unmarried, since each of the bachelors are merely contingently unmarried. In light of this consideration, I will discard the restriction to *objectual* essence throughout the discussion. Still, it is worth noting that all objectual essences *regarding the object itself* obtain essentially of the object in question, in the mediate and consequential sense. For example, suppose $Fx \leftarrow_x x = a$. This yields $E_{\lambda y.y=a} \forall x((\lambda y.y = a)x \supset Fx)$; and by β-reduction $E_{\lambda y.y=a} \forall x(x = a \supset Fx)$. Finally, by the rule of identity, we can infer $E_{\lambda y.y=a} Fa$, which implies εFa.

2 The gap challenge

Obviously, any essentialist account should not take *to obtain essentially* to be understood in terms of *to be necessary*. For example, suppose that for Fa to obtain essentially of a just is for a to necessarily have F. Then to say the necessity of Fa is explained by $E_a Fa$ just amounts to saying that Fa is necessary because a necessarily has F. In this case, essentialism results in blatant circularity.

On the other hand, it is entailed by the thesis of essentialism for any p, εp grounds $\Box p$. But then one might ask how, if essence should not be understood in terms of modality, i.e., has a non-modal nature, one can ensure such an intimate connection between *to obtain essentially* and *to be necessary*. The challenge is often raised against the extensional relation between them. The grounding thesis implies what Mackie (2020) calls "Necessity Principle (NP)":

(NP) $\forall p(\varepsilon p \supset \Box p)$.

and various candidates for counterexamples to (NP), and thus to essentialism, have been suggested. For example, it is pointed out that some facts about persistence and destruction conditions of a particular object, facts about individuation or falling under a sortal are contingent (Leech, 2018; Mackie, 2020). And if an essence fact about an object—which obtains essentially—is roughly a fact about what the object is, plausibly those contingent facts are facts obtaining essentially.

Essentialists, of course, can reply to the alleged counterexamples one-by-one, by arguing that there are reasons why each contingent fact is not something obtaining essentially. But challengers often point out that there is a general conceptual problem lurking behind. As mentioned before, essentialists should not understand what it is to obtain essentially (viz., "ε") in terms of a fact being necessary. On the other hand, if there is no modal import at all in what it is to obtain essentially, we cannot ensure that all facts obtaining essentially are necessary, let alone one being grounded in another. But it is impossible to say that essence is both non-modal and modal. As Mackie asks, how can essentialists "deliver a modal rabbit out of a non-modal hat"? (2020, 252)

One strategy essentialists *cannot* adopt is to insist that (NP) simply is a fundamental fact of reality, and thus refuse to give a further explanation of (NP).[8] First, since (NP) involves a universal quantification, it is plausibly grounded in each instance of p, say q, satisfying $\varepsilon q \supset \Box q$, which is plausibly further grounded in either $\neg\varepsilon q$ or $\Box q$. Second, essentialism implies that *to be necessary* is grounded, and therefore nonfundamental; whereas if (NP) is fundamental, assuming the principle of purity, any constituent of (NP) including *to be necessary* (viz., "$\Box$") would also be fundamental, thus contradiction.

Preexisting attempts by essentialists to support (NP) by their preferred understanding of what it is to obtain essentially have faced strong objections. One common approach among essentialists is to understand essence in terms of *real definition*. Roughly, it is the thesis that the constitutive and immediate essence of something is captured by its real definition (e.g., Fine,

[8] I assume that only ungrounded facts can be fundamental, that is, the 'minimalist' worldview of Fine (2001).

1994; Lowe, 2012; Hale, 2013). Hence, if $\Phi\alpha$ obtains essentially of α, from $\neg\Phi\beta$ we can infer that $\alpha \neq \beta$ for β fails to satisfy the definition of what it is to be identical to α. Hale tried to argue that this conception of essence already grants (NP). Again, let $\Phi\alpha$ obtain essentially of α. Now suppose for contradiction that it is possible that $\neg\Phi\alpha$. It implies that there is a β such that $\alpha = \beta \wedge \neg\Box\Phi\beta$. But β cannot be identical to α in a world where $\neg\Phi\beta$, for it does not satisfy the condition to be identical to α, which is Φ. Therefore, by the necessity of identity $\neg(\alpha = \beta)$, contradiction. Finally, by *reductio* we can conclude it is not possible that $\neg\Phi\alpha$. In other words, if $E_{\lambda x.x=\alpha}\Phi\alpha$, then $\Box\Phi\alpha$ (Hale, 2013, 133).

However, it has been pointed out that the definitional conception of essence cannot guarantee (NP), at least in the way Hale proposed (Romero, 2019; Leech, 2021). For the proposed argument to succeed, it must be assumed that Φ is *necessarily* the definition of α. That is, either $\Box E_\alpha\Phi\alpha$ or $\Box\Phi\alpha$ is required. But what is assumed is only that Φ is *actually* the condition for identity, whereas to further assume $E_\alpha\Phi\alpha$ implies $\Box\Phi\alpha$ simply begs the question.[9]

Alternatively, Correia and Skiles (2017) suggested that essence should be understood in terms of generalized identity. Let "$\phi \equiv \psi$" mean that ϕ is generically identical to ψ. Then q is a full essence of p iff $p \equiv q$, and r is a partial essence of p iff there is a q such that $p \equiv (r \wedge q)$ (2017, 649–50). To understand essence in terms of generalized identity may seem to offer better prospects for advocating essentialism,[10] since "[a]s with objectual identity, generalized identity is tightly linked to metaphysical necessity" (2017, 656). Here $p \equiv p$ is a theorem,[11] it follows in any normal modal logic that $\Box(p \equiv p)$. Then, given $p \equiv (q \wedge r)$, which means that q is a partial essence of p, by substitutability we get $\Box(p \equiv (q \wedge r))$. That is, assuming q is a partial essence of p, it follows that q is a partial essence of p *necessarily*.

This actually proves something stronger than (NP): namely, $\varepsilon p \supset \Box\varepsilon p$. However, since whatever obtains essentially of something also obtains *simpliciter*, this consequence implies (NP). The problem here is that the proof presupposes $\Box(p \equiv p)$: that generalized identities

[9] Of course, Hale himself would reply that the necessity of $E_\alpha\Phi\alpha$ is not to be explained by further realized essence facts. Rather, it is a fundamental necessity that does not permit further explanation (Hale, 2013, 158). However, this move, whether it is convincing or not, is not open to essentialists in our sense.

[10] As far as I know, there are no actual essentialists in our narrow sense who appeal to such a conception of essence. The allegedly failed argument below is rather constructed by challengers of essentialism (Romero, 2019; Leech, 2021).

[11] Since they take generalized identity to be reflexive (2017, 645).

are necessary. Then what could be the source of the necessity of $(p \equiv p)$? It has been argued that the necessity of $p \equiv p$ cannot be explained by the essence of what it is to be generically identical (Leech, 2021, 901–2; see also Romero, 2019, 138). For example, suppose that the necessity is explained by the fact that *for two things to be generically identical partially is to be in a reflexive relation*. But then, what explains the necessity of the partial generalized identity between generalized identity and reflexivity? Here we cannot appeal to the necessity of generalized identity, for that exactly is the supposed explanandum. Therefore, to appeal to the essence of generalized identity does not solve the problem.

Under these considerations, Leech concludes that in explaining the necessity of generalized identity, "*one should not appeal to the essence of anything to account for this necessity*". She further considers this constraint "the crucial and general objection" to the necessity of facts obtaining essentially (2021, 902).

I disagree. Leech overlooks a remaining possibility: that all generalized identities are necessary because of the essence of what it is to be necessary, that is, $\Box$. Furthermore, in my view, this also applies directly to (NP). I agree that any non-modal specification of essence cannot be used to metaphysically guarantee (NP). However, every fact obtaining essentially is necessary, not because of the nature of what it is to obtain essentially, but because of the nature of what it is to be necessary. In the next section, I first argue that essentialists must take (NP) to obtain essentially of $\Box$, and then show how adopting this thesis helps essentialists respond to the gap challenge.

3 The essentialist essence of *what it is to be Necessary*

Essentialists who take (NP) to hold would also wish to take (NP) to hold *necessarily*. There are various possible motivations behind this position. If an essentialist sides with *grounding necessitism*, viz., the thesis that the grounds necessitate the grounded, it would be a natural consequence that (NP) is necessary.[12] But even if one favors *grounding contingentism*, viz.,

[12]It is worth noting that grounding necessitism is extremely natural for essentialists who adopt a unified foundation of essence and ground.

the thesis that the grounds do not necessitate the grounded, one may still take (NP) to hold necessarily in order to avoid conflict with plausible metaphysical assumptions. To reason in terms of modal logic without giving metaphysical import: if (NP) is not necessary (in the actual world @), there is a possible world accessible from @ where $\exists p(\varepsilon p \wedge \neg\Box p)$. And under essentialism, the witness of p, say q, either does not obtain essentially in @, or it does and therefore $\Box q$ holds in @. In the latter case, we straightforwardly get a counterexample to the S4 principle: $\Box\phi \supset \Box\Box\phi$, which is commonly thought to hold for metaphysical modality. In the former case, since essentialists also hold that $\varepsilon\phi \supset \Box\varepsilon\phi$[13], by S4 it follows that all realized essence facts obtaining in @ obtain in all possible worlds; but in some non-actual possible worlds there obtain additional realized essence facts. However, there does not seem to be any good reason to think that the actual world has the fewest realized essence facts among all possible worlds, if there can be a discrepancy in realized essence facts between worlds at all.

Thus, in any case, essentialists would admit that (NP) is necessary. Moreover, given essentialism, to say that (NP) is necessary amounts to saying that (NP) obtains essentially of something. Yet of what, if anything, would it obtain essentially?

It would be promising to search for the subject of which (NP) obtains essentially by inspecting (NP) itself. Notice that the formalization of (NP) comprises a universal quantification over facts, ε, a material conditional, and $\Box$. But it should be obvious that (NP) would have nothing to do with the nature of the material conditional, so we are left with three candidates. It might seem equally farfetched to think that (NP) obtains essentially of the logical entity which is associated with the universal quantifier over facts. It might sound better to say that (NP) obtains essentially of *all facts*. The general idea might be that, for each individual fact q, it obtains essentially of q that $\varepsilon q \supset \Box q$; and if it is somehow given that the totality of facts contains *all* facts, (NP) would obtain essentially of the totality of facts.

But could (NP) obtain essentially of yet something else? To be sure, trivially, if (NP) obtains essentially of one of the three candidates mentioned in the previous paragraph, (NP) will also obtain essentially—in the mediate sense—of anything whose essence involves it. But let us exclude such cases from consideration due to their metaphysical insignificancy. Still, one pos-

[13]Immediate from $E_F\phi \supset E_F E_F\phi$ and (NP).

sible line of thought is that although (NP) is not itself fundamental, it might obtain essentially of fundamental metaphysical laws, or something similar.

In sum, we are left with four options: (NP) obtains essentially either of ε, of all facts, of some fundamental metaphysical laws, or of $\Box$. I will argue in detail that every option except the last fails, and then show how the final option helps to answer the gap challenge on behalf of essentialists.

3.1 Taking (NP) to obtain essentially of ε

It is worth noting that it has already been suggested by Wilsch (2017) and Wallner and Vaidya (2020) that (NP) or something stronger obtains essentially of ε, although Wilsch is not an essentialist in our sense.[14] Wilsch directly takes (NP) to obtain essentially of ε, whereas Wallner and Vaidya take both $\forall p(\varepsilon p \supset \Box \varepsilon p)$ and $(\varepsilon p \textit{ can explain } \Box p)$ to obtain essentially of ε. Since $\varepsilon p \supset \Box \varepsilon p$ entails (NP), it would be fair to say both take (NP) to obtain essentially of ε. Following Wilsch, let us term the thesis that (NP) obtains essentially of ε "Modal Essence of Essence", "(MEE)" for short. That is:

(MEE) $E_\varepsilon \forall p(\varepsilon p \supset \Box p)$.

In response, Romero (2023) argues that (MEE) results in the claim that $p \wedge q$ grounds p, which infringes the irreflexivity of ground.[15] His argument is convincing if one adopts the conception of essence in terms of generalized identity. To get (MEE), suppose that $\varepsilon p \equiv_p (\Box p \wedge q)$—that is, $\Box$ is partially essential to ε. This yields $E_\varepsilon \forall p(\varepsilon p \supset (\Box p \wedge q))$, and consequentially $E_\varepsilon \forall p(\varepsilon p \supset \Box p)$. Now, since two generically identical statements would be substitutable even in some hyperintensional contexts (Dorr, 2016; Rayo, 2013), if p grounds q and $p \equiv r$, r should

[14] Wallner and Vaidya explicitly state that "modality is grounded in definitional essences." (2020, 431). By contrast, Wilsch explicitly describes himself as a modal primitivist. Although he takes essence to give the "source" of metaphysical necessity, here "[t]he sources of necessity are the modal joints of nature." (2017, 429), which determine the extension of various kinds of necessity and account for their naturalness. Essence, then, explains why metaphysical necessity is a significant kind of primitive necessity, why it "carves the nature at its joints". This is clearly a different enterprise from the essentialism on the source of necessity being discussed in this paper.

[15] To be precise, he appeals to the irreflexivity of *metaphysical explanation* instead of grounding, since Wilsch is not committed to the thesis that essence grounds necessity. Here I focus on grounding to keep the discussion compact.

also ground q. To illustrate: suppose x's not being eligible for a marriage loan is grounded in x's being a bachelor. Since to be a bachelor just is to be an unmarried man, x's not being eligible for a marriage loan should also be grounded in x's being an unmarried man. Similarly, given $\varepsilon p \equiv_p (\Box p \wedge q)$, the essentialist thesis that εp grounds $\Box p$ is equivalent to the claim that $\Box p \wedge q$ grounds $\Box p$. And that is evidently not satisfactory; therefore, (MEE) is inconsistent with essentialism.

However, it is not obvious whether this argument works without Correia and Skiles' conception of essence. The argument crucially relies on the premise that if q is the full essence of p, "q" must be able to replace "p" in grounding contexts. This is not very obvious for two reasons. First, it is far from clear that there is a metaphysically significant notion associated with "full essence", if essence is not to be understood in terms of generalized identity. Second, even if we grant the notion of full essence, one can still contend that the full essence of something is intimately related to, yet not (generically) identical to it. Then one would at least require a substantial theory of essence and ground to warrant the claim that if p grounds q and $p \equiv r$, r must also ground q.

But there is a general argument for the incompatibility of essentialism and (MEE). Let us first introduce the notion of essential dependence: x essentially depends on y, in Fine's terms, "if y is a constituent of a proposition that is true in virtue of the identity of x" (Fine, 1995b, 275; the term "essential dependence" follows Correia (2008)). Therefore, from $E_\varepsilon \forall p(\varepsilon p \supset \Box p)$, it follows that ε depends on $\Box$. By contrast, since standard understandings of ground (Schaffer, 2009; Fine, 2012) take the grounds to be strictly prior to, or more fundamental than the grounded, from essentialism—$\varepsilon p \rightarrow_p \Box p$—it follows that ε is prior to $\Box$, which implies that ε cannot depend on $\Box$.[16] That is, while it follows from (MEE) that ε depends on $\Box$, from essentialism it follows that ε cannot depend on $\Box$. Therefore, (MEE) is incompatible with essentialism.

The appeal to essential dependence may seem somewhat awkward, for I am operating with

[16]Following Fine (1995b), I use the term "dependency" for the fundamentality relation that does *not* imply antisymmetry (1995b, 282–3). If X depends on Y, one can only infer that X is *not prior* to Y, for the possibility of Y being dependent on X is still open. The term "prior to" is reserved for the antisymmetric fundamentality relation.

the consequential sense of essence, whereas the notion of essential dependence is usually formulated in terms of constitutive essence for good reasons. For example, assume the standard definition of the biconditional "$\Leftrightarrow$" can also be read as a generic grounding claim: $((p \supset q) \wedge (q \supset p)) \rightarrow_{p,q} (p \Leftrightarrow q)$. The obvious implication would be that the higher-order relation $\supset$ is strictly prior to the higher-order relation $\Leftrightarrow$. Yet since $p \Leftrightarrow p$ is a logical truth, it would be a consequential essence of anything, including $\supset$. But if $(p \Leftrightarrow p) \leftarrow_{p,q} (p \supset q)$, we get $E_{\supset}\forall p \forall q((p \supset q) \supset (p \Leftrightarrow p))$, and therefore $\supset$ depends on $\Leftrightarrow$, thus contradiction.[17]

Now, if (NP) is proposed as obtaining constitutively essentially of ε, the aforementioned worry does not arise, and my argument remains intact. Moreover, sometimes even if (NP) obtains consequentially essentially, the validity of the argument is not affected. For example, essentialists might wish to say it is rather the biconditional $\forall p(\varepsilon p \Leftrightarrow \Box p)$—which immediately implies (NP)—that obtains constitutively essentially of ε, whereas (NP) is merely consequential. I find this proposal reasonable. Nevertheless, my argument *still* retains its force, for "$\Box$" occurs in the biconditional proposition as well.

So we must examine *how* (NP) would obtain consequentially essentially. That is: what should obtain constitutively essentially of ε, so that (NP) logically follows from them? We can immediately list trivial cases: (NP) would obtain consequentially essentially if (1) (NP) follows from logic alone; and (2) the set of facts obtaining essentially of *to obtain essentially* is inconsistent. Also, since (NP) contains a material conditional, by its logic (NP) could obtain consequentially essentially if: (3) $\forall p \neg \varepsilon p$ follows from what obtains constitutively essentially; and (4) $\forall p \Box p$ follows from what obtains constitutively essentially. Otherwise, at least in standard modal logics, to prove (NP) the premises must also have modal components. Thus, (NP) could obtain consequentially essentially only if: (5) the constitutive essence includes $\Box$; (6) the constitutive essence includes $\Diamond$. (1)–(6), although not mutually exclusive, would be exhaus-

[17] Fine (1995b) tries to solve this problem by introducing the additional condition: "*x* depends upon *y* just in case *y* cannot be generalized out of the consequentialist essence of *x*" (1995b, 278), where *y* can be *generalized out* of the essence of something if all occurrences of *y* in the essence can be replaced by any object and still belong to the essence. This can perhaps solve the problem regarding the dependence between first-order objects. Nevertheless, it does not seem to work regarding dependence outside first-order objects. What I proposed seems to be a counterexample: $\Leftrightarrow$ in the consequential essence of $\supset$ cannot be generalized out, for substituting some two-place higher-order relation for $\Leftrightarrow$ would not result in logical truths. See also **?**.

tive.[18]

(1)–(4) are untenable, whereas (5) and (6) show that ε still depends on $\Box$ or something similar. As for (1), it is clear that (NP) does not follow from some customary logics—classical first-order logic, normal modal logics, and so on. If (NP) were a logical fact in that sense, the gap challenge would not arise at all, for we do not generally ask for an explanation of a logical truth even if some non-logical explanation can be given. Of course, we might construct a logic where (NP) follows logically. But then we would be confronted with the philosophical problem of why we should adopt such a logic, and in all likelihood the philosophical motivation would be essentialism itself. Then the opponent can raise the gap challenge, which will bring us back here once again. Therefore, one cannot appeal to (1) to escape the priority problem.

(2). Since what obtains essentially obtains *simpliciter*, endorsing (2) amounts to taking the world to be inconsistent. This is presumably not a price many philosophers want to pay, and those who are willing to accept paraconsistency would typically refuse to say we can infer everything from inconsistent premises. Regarding (3), together with essentialism, this means that there are no sources of necessity, and therefore no necessary facts. It is intuitively not appealing, and the task of making sense of the proposal and refuting it would not be adequate for the purpose of the paper. At least dialectically, I expect one who enters the debate over the source of necessity would not hold that there are no necessary facts. The same consideration applies to (4), which means that all facts are necessary.

(5). It would be obvious that if the constitutive essence of ε includes $\Box$, it follows that ε indeed depends upon $\Box$. And as I argued, this undermines the initial essentialist account. To show that (6) still results in undesirable consequences requires more care. One easy way out, of course, is to take $\Diamond$ to be defined in terms of $\Box$, so that you cannot get (6) without (5). But perhaps one would prefer to take $\Box p$ to be defined as $\neg\Diamond\neg p$, or to take the two operators to be on a par in regard to fundamentality. Throughout the paper, I have implicitly assumed that $\Box$ is the fundamental modal operator, and essentialism is an account of the source of *necessity*, not other kinds of metaphysical modality. However, this assumption has been made only because

[18] I have especially classical S5 logic in mind; but it is also exhaustive even if one endorses S4 or intuitionistic logic.

it facilitates the discussion. It is certainly possible for the essentialist account to be formulated in terms of possibility: the *possibility* of any possible fact q is explained by $\neg\varepsilon\neg q$. If $\Diamond$ is more fundamental than $\Box$, essentialism probably should undergo such a modification. But once the modification is made, (6) undermines the account, for it implies that ε is dependent upon $\Diamond$, that is, *to be possible*.

Hence, endorsing (MEE) would in any case make ε essentially dependent on $\Box$. Still, it might be argued that this does not preclude ε from grounding $\Box$. The reasoning for the preclusion runs as follows. (i) If F essentially depends on G, F is not prior to G; (ii) If F grounds G, F is prior to G. Conclusion: if F essentially depends on G, F cannot ground G. The worry, however, is that for the inference to be valid, "prior to" in (i) and (ii) must express the same metaphysical relation, which can be contested (Vogt, 2026).

However, to draw the conclusion, one need not hold that (i) and (ii) operate with the identical notion of priority. To show that two priority relations cannot head in opposite directions—where F is prior to G in one sense and G is prior to F in the other sense—would suffice. This is achieved, for example, if we show that both grounding and essence give constitutive conditions, where presumably constitutiveness cannot be circular (Fine, 2015). Also, one can argue that real definition—which is sometimes taken to be the constitutive and immediate essence—of whatever can also be formulated in terms of grounding (Rosen, 2015, 198–9). I am inclined to think that the account of Fine (2015) is largely correct, but to argue for the macroscopic metametaphysical theory would certainly exceed the scope of the paper. Still, some direct arguments for the opposite-heading of priorities should be dealt with.

Vogt suggests that, assuming the Aristotelian account of property, there can be cases where E_aPa but Pa grounds *the property P exists* (2026, 519). Vogt seems to regard this as a case in which a is both prior (by grounding) and not prior (by essential dependence) to P. However, one should note that "P" is used on one occasion as a name for a first-order object, and on the other as a predicate. Thus, what the grounding statement and the essence statement jointly imply is actually that a is not prior (by essential dependence) to P, while a is prior (by grounding) to the *property P*. This is clearly not a case of two priority relations heading in opposite directions.

Vogt (2026); **?** also cites various specific cases where one might contend that $Fx \rightarrow_x Gx$ but $E_F \forall x(Fx \supset Gx)$ (see also Correia (2018)). Of course, she is cautious enough not to declare any of the particular examples to be definitive. However, it is suggested that the candidate examples, taken together, have enough force to make one take the possibility of opposite-heading seriously. I disagree, for it seems that, even if the grounding claim and the essentiality of the conditional are correct, those putative examples are subject to a uniform treatment. That is: the conditional should obtain essentially of the grounded, not the ground. For example, Vogt holds in both papers that *a promises to p* grounds *a has an obligation to p*, whereas it seems that it obtains essentially of *what it is to promise* that (*one promises to p* $\supset$ *one has an obligation to p*). One of course may deny that there is such an essential connection between *to promise* and *to have an obligation*. But if there is, my treatment is rather to say that it obtains essentially of *what it is to have an obligation* that (one promises that $p \supset$ one has an obligation to p). In other words, it would do equally well to say the conditional obtains essentially of the grounded, not the ground.[19][20]

Therefore, insofar as essentialists seek to ground $\Box$—or $\Diamond$—in ε, they should not think that (NP) obtains in virtue of the nature of ε, viz., *what it is to obtain essentially*.

3.2 Taking (NP) to obtain essentially of fundamental laws, or of all facts

Similar considerations lead us to reject the suggestion that (NP) obtains essentially of some fundamental laws, that is:

(MEL) [21] $E_L \forall p(\varepsilon p \supset \Box p)$, where L is the (totality of) laws of which (NP) allegedly obtains essentially.

[19] This is exactly parallel to the goal of this section. Essentialists would agree that εp grounds $\Box p$. But if one also has to think that the conditional $\varepsilon p \supset \Box p$ should be essential, then it should be essential to the grounded (viz., $\Box$), not to the ground (viz., ε).

[20] One might criticize such a treatment as *ad hoc*. It is perhaps so in some sense, but notice this does not increase the theoretical complexity of priority, but reduces it. If the priority by grounding and the priority by essential dependence were so fundamentally different that opposite-heading would be permitted, it seems that there should be two fundamentally different accounts of what metaphysical priority is, perhaps even two different kinds of fundamental reality, which would have to be altogether compatible. In the absence of such an account, I regard my move as justified.

[21] Modal Essence of Laws.

Recall that I argued in §2 that (NP) is not fundamental. Then we can again appeal to the notion of essential dependence, and argue that (MEL) implies that L depends on (NP), and therefore (NP) is at least as fundamental as those laws, i.e., fundamental. This contradicts the observation that (NP) is not fundamental, especially if essentialism holds.

Again, the argument in the previous paragraph is awkward since it operates with the consequential notion of essence. Still, the list (1)–(6) is exhaustive even if we replace the subject of which (NP) obtains essentially with the fundamental laws in question. Then we can equally well establish that (MEL) implies that the fundamental laws depend on metaphysical modality.

Consideration of essential dependence likewise undermines the idea that (NP) might obtain essentially of *all* facts. According to this suggestion, for each fact p, $\varepsilon p \supset \Box p$ obtains essentially of p. Now, let q be an instance of p that is a fundamental fact. By essential dependence, it follows that $\Box$ is fundamental, whereas by essentialism—$\varepsilon p \rightarrow_p \Box p$—it follows that $\Box$ is not fundamental.

One might object that it is unfair to make the assumption that if (NP) obtains essentially of all facts, every particular fact should have necessity-related essence. The problem I raised will not arise if only some nonfundamental facts have a necessity-related essence, and these limited cases are sufficient to grant that (NP) obtains essentially of the totality of facts. I find such scenarios implausible. In those scenarios, not only would selected nonfundamental facts guarantee, by their nature, that if numerous facts obtain essentially then they are necessary; they would also have to guarantee that those numerous facts are *all the facts there are*. I see no plausible way in which the latter could be secured.[22]

But there is a final suggestion worth discussing. Instead of appealing to the essence of *all facts*, we may appeal to the essence of *what it is to be a fact*. That is:

(MEF) [23] $(\varepsilon X \supset \Box X) \leftarrow_X \mathit{Fact}(X)$.

(MEF) does not directly grant E_{Fact}(NP), but rather $E_{\mathit{Fact}} \forall X (\mathit{Fact}(X) \supset (\varepsilon X \supset \Box X))$. But since

[22] It is indeed possible that (NP) is *essential* to some nonfundamental facts. In fact, it is plausible that (NP) is essential to (NP) itself. But from this, we cannot infer that (NP) *obtains essentially* of (NP).

[23] Modal Essence of Facts.

it is extremely plausible that either $E_{Fact}\forall p\ Fact(p)$ or $E_{\forall_t}\forall_t p\ Fact(p)$ where $\forall_t$ is specifically the quantifier quantifying over facts, it follows that with (MEF) E_{Fact}(NP) or $E_{\forall_t}$(NP) holds.

However, (MEF) is not metaphysically convincing. Intuitively, if the primary subjects of being necessary and obtaining essentially are facts, *to be a fact* should be prior to $\Box$ and ε. Among a variety of facts, only a proper subclass of them would obtain essentially (and therefore necessary, given essentialism). I submit that this gives at least a *prima facie* reason to think *to be a fact* does not depend on ε, especially if essentialists hold that a *fact*'s being necessary is grounded in a *fact*'s obtaining essentially. To draw an analogy, it is more plausible to think *to be a natural number* is prior to *to be prime*, especially if we take it that a *natural number*'s being prime is grounded in the *natural number*'s not being a product of two natural numbers less than itself. At least, one would hardly maintain that it is essential to *what it is to be a natural number* that a natural number is prime if it is not a product of two natural numbers less than itself.

3.3 Taking (NP) to obtain essentially of $\Box$

The only remaining possibility, then, is to say that (NP) obtains essentially of $\Box$, viz.:

(EEN) [24] $E_{\Box}\forall p(\varepsilon p \supset \Box p)$

This is what I wish to advocate, provided that essentialism is correct. It should be clear that, in this case, there is no problem with regard to fundamentality. From (EEN) it follows that $\Box$ is not prior to ε, which is harmonious with the grounding claim that εp grounds $\Box p$.

Aside from this metametaphysical harmony, we should also examine whether (EEN) is metaphysically reasonable. One reason to endorse (EEN), of course, stems from the plausibility of essentialism. We established that essentialists should take (NP) to obtain essentially, and other plausible candidates for the subject of which (NP) obtains essentially fail. Thus, the only remaining way to maintain essentialism is to endorse (EEN). And since essentialism entails (EEN), it must be at least as reasonable as essentialism. I will not explore the theoretical moti-

[24] Essentialist Essence of Necessity.

vations for essentialism here, for I do not aim in this paper to argue for the essentialist answer against other alternatives. Yet those motivations support (EEN) as well.

There is another way to approach (EEN) through Lowe (2008)'s slogan that *essence precedes existence*. The idea, as I conceive it, is as follows. It should be obvious that if Xs exist, it is sensible to speak of what it is to be X. But even if we somehow discovered that Xs do not exist, to ask whether Xs exist, or even to deny their existence[25] requires that we at least partially know the nature of what it is to be X (see also Fine, 2017). Thus, if the denial of the existence of Xs is sensible, there should be knowable facts regarding the nature of what it is to be X, regardless of the existence of Xs. At least, we cannot deny that p is essential to what it is to be X merely on the basis of the nonexistence of Xs. And if what it is to be X has an essence, there must also be realized essence facts regarding it.

It is common to think that there are varieties of necessity, regardless of whether there is a single fundamental kind of necessity from which any other necessities are derivative. Given *essence precedes existence*, we can posit a realized essence facts for each kind of necessities without committed to its fundamentality or existence in an ontologically loaded sense. For example, it is plausible that given the nomic necessity operator $\Box^N$ and the predicate "*NL*" for "*to be a fact of natural laws*", $E_{\Box^N}\forall p(NL(p) \supset \Box^N p)$. Now suppose we carve out a new kind of necessity: the *essentialist necessity*. Let "$\Box^E$" be the essentialist necessity operator. Suppose further that part of the nature of *to be essentialistically necessary* is captured by (EEN)^{E}: $E_{\Box^E}\forall p(\varepsilon p \supset \Box^E p)$. If the realized essence fact of $\Box^N$ is plausible, so is (EEN)^{E}. And one way to derive (EEN) from (EEN)^{E} is to argue that the metaphysical necessity ("necessity" for short throughout the paper) just is what we previously called essentialist necessity.[26]

Again, to argue for the equivalence between two kinds of necessity would require another twelve or more pages, and definitely exceed the scope of this paper. Still, the tenability of (EEN)^{E} at least indirectly shows that (EEN) is not metaphysically implausible in the way that (MEF) is. If one can acknowledge (EEN)^{E} but not (EEN), one should provide reasons why

[25] with ontological significance, or in so-called "ontologese" (Sider, 2009; see also Dorr, 2005).

[26] The general idea of this scheme is drawn from reading Fine (2020, 463–4), although I am not confident whether this reflects Fine's intention. At least Romero (2023) seems to interpret it in the opposite way, where Fine is taken to be committed to something similar to (MEE).

metaphysical necessity is crucially different from essentialist necessity.[27] And there seems to be no quick argument for it. At least on the surface level, there are abundant similarities between them: both take facts to be their subject, both are alethic necessities, both are non-epistemic, and so on.

And I further argue that essentialists can answer the gap challenge by appealing to (EEN). To briefly reiterate, their main criticism was that once we stick to the notion of essence that is *not* understood in terms of necessity, there seems to be a conceptual gap between essence and necessity. I agree with Romero (2019) that essentialists cannot take facts obtaining essentially to be *ab initio* necessary. Nevertheless, there are many philosophers with the essentialist intuition that it is in some sense "obvious" that facts obtaining essentially are necessary. There certainly is some kind of frustration when essentialists are confronted with questions like "why are facts obtaining essentially necessary" or "why does something necessarily have the property[28] that defines what it is?" This is, I suggest, the same kind of frustration when one is confronted with questions like "why is water H_2O?" or "why does Socrates' singleton set have Socrates as a member?". As Glazier (2017) argues, we can simply answer: "that is just what water is" or "that is just what the set is." Turning to the question "Why are facts obtaining essentially necessary?", I suggest that the answer essentialists can give is: "that is just what $\Box$ is." And to give such an answer, one must first accept (EEN).

Thus, although essentialists must endorse (EEN) for an independent reason, this simultaneously provides a viable reply to the gap challenge. Essentialists can reply that it is misleading to ask whether ε has a suitable nature to be the source of necessity. Rather, it is the nature of $\Box$ that is intimately related to ε, and conceptually connects $\Box$ and ε. Mackie (2020) is correct to point out that essentialists cannot deliver a modal rabbit (the *necessity* of facts obtaining essentially) out of a non-modal hat (a non-modal conception of essence). Yet essentialists simply do not have to. What they have to do, and are able to do, is to deliver an essentialist rabbit (the necessity of facts *obtaining essentially*) out of an essentialist hat (an essentialist conception of metaphysical necessity).

[27] Notice that although I argue that (EEN) is forced upon essentialists, the converse does not hold. For example, the thesis that both necessity and essentiality are fundamental is compatible with (EEN).

[28] Here "property" is not to be understood necessarily as a term for first-order objects.

Furthermore, certain relations between essence and necessity assumed by essentialists to vindicate (NP) can be inferred from (EEN). For example, let us revisit Hale's assumption that given $E_\alpha\Phi\alpha$, $\Diamond\neg\Phi\beta \supset \alpha \neq \beta$. Suppose for contradiction that $\alpha = \beta$ and $\Diamond\neg\Phi\beta$ ($\equiv \neg\Box\Phi\beta$). By substitutivity, it follows that $\neg\Box\Phi\alpha$. Now, from (EEN) we infer $\forall p(\varepsilon p \supset \Box p)$, and thus $E_\alpha\Phi\alpha \supset \Box\Phi\alpha$. Finally, by the assumption $E_\alpha\Phi\alpha$ and *modus ponens* we derive $\Box\Phi\alpha$, which is a contradiction. Therefore, there can be no β such that $\alpha = \beta \wedge \neg\Box\Phi\beta$, which ensures that if $\Phi\alpha$ obtains essentially of α, then Φ is a metaphysically necessary condition for something to be identical to α.

A similar moral applies to the necessity of generalized identity. Although (EEN) is sufficient to secure the necessity of generalized identity,[29] if one is sympathetic to Correia and Skiles (2017), in the same spirit as (EEN), one can instead adopt $E_\Box((\phi \equiv \psi) \supset \Box(\phi \equiv \psi))$. That is, $\Box$ is essentially such that if a generalized identity actually holds, it holds necessarily. It is again a mistake to assume that the source of the necessity of generalized identity must be found in the essence of *to be generically identical to*, or what *to be generically identical to* is generically identical to. Instead, essentialists can simply answer that all generalized identities are necessary because that is just what $\Box$ is.

4 Circularity

I argued in §3 that essentialists should accept (EEN). This was done by showing that any alternative is incompatible with essentialism. But to say that it is the only viable option open to essentialism is not to say that it is tenable. After all, essentialism might also be incompatible with (EEN); and together with the argument I have presented, such incompatibility would complete a trilemma that undermines essentialism.

It seems clear that (EEN) is not *logically* inconsistent with essentialism. To say what it is to be necessary is such-and-such is one thing, and to say that necessity is grounded in something is another. And I also argued in §3.3 that (EEN) by itself is not metaphysically awkward, as was

[29] Given that $p \equiv q$ indicates that q is essential to p, since the fact that q is essential to p obtains essentially of p, we can infer that $E_p(p \equiv q)$. And since by (EEN) $\Box$ is such that whatever obtains essentially is necessary, it follows that $p \equiv q$ is necessary.

the case with (MEF). But it may be the case that the thesis seriously undermines the theoretical motivation of essentialism. The reason for rejecting the thesis that (NP) obtains essentially of *what it is to obtain essentially* was not logical inconsistency. The argument against it was that it delivers a verdict on priority which is inconsistent with essentialism.

As I argued in §3.3, (EEN) does not face the same problem. But there perhaps are other difficulties. Romero (2019, 124) states six constraints that an explanation of modality by essence must satisfy:

Why The explanation must be an answer to a 'why' question.

Acyclicity The explanation should not be circular.

Fundamentality The explanans should be metaphysically more fundamental than the explanandum.

Ontic and Factive It must be the case that the truth of the explanans, not our concept or theory of it, explains the truth of the explanandum.

Structural The explanandum need not be an object or a property.

Objectivity The explanation should be objectively valid.

It has already been shown that (EEN) does not cause essentialism to violate **Fundamentality**. Also, essentialism with or without (EEN) easily satisfies both **Why** and **Ontic and Factive**. It answers the question "*Why* is *p* necessary?" by appealing to the realized essence *facts*, not our concept of essence. Essentialists would certainly hold that if essentialism on the source of necessity is valid, then it is **objectively** valid. Finally, **Structural** is not a negative constraint that one can infringe.

It is perhaps less obvious, however, whether essentialism with (EEN) is compatible with **Acyclicity**. Some might find the idea that the nature of $\Box$ validates the explanation of $\Box$ in terms of ε suspicious. But if it is indeed circular, exactly which part of the picture is viciously circular?

First, notice that I do not try to explain the necessity of p by appeal to (EEN). The explanatory work is done by the fact that p obtains essentially, and (EEN) is not involved as a source of the necessity of p. However, it is far from clear why including (EEN) as the source would be viciously circular. Suppose I am wrong, and every source of necessity must include (EEN): p is necessary because p obtains essentially *and every such fact is necessary in virtue of the nature of* $\Box$. What is wrong with this kind of explanation? Perhaps my point can be made by a simple analogy. If someone asks me why 3 is a prime number, I can simply answer "because 3 is a natural number that is not a product of two natural numbers smaller than itself." However, if someone thinks this is incomplete, then I can add: "... *and* every such natural number is prime in virtue of the nature of *what it is to be prime*." It should be obvious that neither of the explanations for the fact that 3 is prime is viciously circular.

Perhaps another charge can be pressed by focusing on the particular necessity of (NP). As I argued in §3, essentialists must admit that (NP) is necessary, and that $\Box$(NP) is grounded in (EEN), that is, $E_{\Box}$(NP). This might seem somewhat problematic. One might contend that it is viciously circular to appeal to the nature of $\Box$ in explaining why $\Box$(NP) holds, given that (NP) obtains essentially of $\Box$.

However, I still do not see anything problematic here. To generalize, someone who finds the situation problematic would assume that if ϕ contains X, then $E_X\phi$ cannot explain $X(\phi)$. But this principle is not convincing. For example, suppose a quasi-God ("Q" below) knows all and only the facts that obtain essentially, and its knowing any fact obtaining essentially is metaphysically grounded in that fact's obtaining essentially. Let us further suppose it is essential to *what it is to know* that if someone knows that p, it is the case that p, that is, $E_K\forall p(\exists sK(s,p) \supset p)$. It then follows that $E_K\forall p(K(Q,p) \supset p)$ in the consequential sense. Now, by the assumption regarding Q, it is the case that $K(Q,\forall p(K(Q,p) \supset p))$, and this is moreover partially grounded in $E_K\forall p(K(Q,p) \supset p)$. Clearly, there is nothing viciously circular here. But this is exactly the instance being discussed: namely, that there being a ϕ containing X such that $E_X\phi$, where $E_X\phi$ explains $X(\phi)$.

5 Conclusion

I have argued that essentialists must accept (EEN), and that by accepting (EEN), they can respond to the gap challenge. The challenge points out that essentialists must adopt (NP), yet one cannot guarantee (NP) from the non-modal conception of essence. Independently, I have argued that essentialists must also adopt □(NP), and the only way to account for □(NP) is to accept (EEN). And it turns out that (EEN) can also be utilized to respond to the gap challenge. Indeed, (NP) is not explicable from a non-modal conception of essence; it is explicable, however, from an essentialist conception of metaphysical modality.

What I have *not* argued in this paper is that essentialism is superior to its competitors. My aim in this paper has been defensive: to show that a particular challenge to essentialism can be coherently resolved via what essentialists must endorse anyway. The gap challenge rather helps to clarify what essentialism entails: *If* essentialism is correct, then so is (EEN). Any argument in favor of essentialism is also an argument for (EEN). Furthermore, if one agrees that *essence precedes existence*, it illuminates another argumentative strategy for essentialists—namely, to argue that metaphysical necessity is essentialist necessity. But of course, the contrapositive of the conclusion also holds. If (EEN) is false, so is essentialism; thus any criticism against (EEN) would thereby be a criticism against essentialism. However, fleshing out new arguments for essentialism or addressing possible criticisms of (EEN) would exceed the scope of this paper.

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